Small and medium-sized enterprises' path to sustainable supply chains: exploring the role of supply chain finance and risk management

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Abstract

Purpose – This paper aims to investigate the triangular interconnections among supply chain finance (SCF), supply chain risk management (SCRM) and supply chain sustainability (SCS) within the context of small and medium-sized enterprises (SMEs) under the theoretical foundation of dynamic capabilities view.

Design/methodology/approach – A total of 319 valid data sets were gathered from SMEs in China to evaluate the research model. This study uses partial least square structural equation modelling and necessary condition analysis as the two statistical methodologies for the assessment.

Findings – The findings indicate that SCF positively impacts on both SCRM and SCS, whereas SCRM also positively influences SCS. Furthermore, it has been observed that SCRM partially mediates the connection between SCF and SCS.

Research limitations/implications – The findings contribute to the literature of SCS by empirically validating the direct and mediating impacts of SCRM on SCS.

Practical implications – The results provide valuable insights that can assist SME stakeholders, owners and managers in developing strategies to effectively incorporate SCF and SCRM practices, thereby enhancing SCS performance.

Originality/value – This study expands the existing research on SCF and SCRM in the context of promoting SCS, specifically from the viewpoint of an Asian developing country.

Keywords Sustainable supply chain, Supply chain finance, Supply chain risk management, Dynamic capabilities view, Necessary condition analysis

Paper type Research paper

1. Introduction

Adopting sustainability principles in the supply chain is one of the most consistent and increasingly important topics for both industry participants and academics. Achieving desirable supply chain sustainability (SCS) satisfies the strong demands of various stakeholders by placing greater emphasis on superior long-term sustainable performance (Arora *et al.*, 2020), particularly in today's hypercompetitive and complicated business landscape (Qorri *et al.*, 2018). The intense demand from industry participants translates into significant drivers and interests for academics to explore SCS towards better understanding from various perspectives. A more comprehensive understanding of SCS's advantages provided by scholars encourages firms to emphasise the adoption of SCS principles.

In the context of small and medium-sized enterprises (SMEs), limited resources are a prominent characteristic; thus, the comprehension and application of resources should be

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highly valued. Considering resources, some researchers have argued that the use of the dynamic capabilities view, an extension of the resource-based view, is relevant for managers to reorganise resources easily to satisfy consumer demand, which is the most critical factor in achieving SCS (Teece *et al.*, 1997). The resource-based view or the dynamic capabilities view have been applied by researchers to explore potential approaches to SCS, such as supply chain risk management (SCRM) (Nisar *et al.*, 2022), supply chain finance (SCF) (Wang *et al.*, 2023), supply chain members' relationships (Kumar and Rahman, 2016) and technological adoption (Hazen *et al.*, 2016).

Adopting sustainability principles in the supply chain is increasingly favoured by large-scale firms (e.g. HP, IBM, BMW and Walmart) to pursue long-term performance (Qorri *et al.*, 2018). However, SMEs lag behind large enterprises and are often criticised for their short operating histories, poor financial indicators, risky investments and inefficient resource usage, which may be potential barriers (Song *et al.*, 2018; Silva *et al.*, 2021). In China's dynamic business landscape, SMEs are the engines of economic growth, comprising more than 90% of all market firms, accounting for 80% of employment, contributing to 60% of the country's gross domestic product and making up 50% of all national tax receipts, with strong political support from the government (Hänle *et al.*, 2022). The SME Development Index indicates that Chinese SMEs have faced development bottlenecks over the past decade and have not performed optimally in the long term (CASME, 2023). Thus, capital shortages and risk management are recognised as the two main problems that must be addressed in this study to achieve SCS by SMEs from the perspectives of SCF and SCRM.

Earlier research posited that fostering SCF can effectively enhance SCS in a large-firm context. This alignment is rooted in the shared objective of prioritising "doing well by doing good" (Toktaş-Palut, 2022). For instance, Zhan *et al.* (2018) explored the payment terms of SCF, indicating that reducing the payment lead-time through SCF practices can encourage SCS and supply chain efficiency. Moreover, some scholars provide empirical evidence in the context of major companies, demonstrating that the cost-effectiveness of SCF can reduce a firm's consumption, thereby ensuring supply chain efficiency in line with sustainable development principles (Wang *et al.*, 2023; Liu *et al.*, 2020). However, prior investigations have found that SCF is fundamentally important for SCRM because it minimises financial risks and improves the dependability of information among companies of different sizes (Beka Be Nguema *et al.*, 2022). Lam and Zhan (2021) revealed that SCF reduces supply chain risks by minimising information disparities regarding financing concerns in public firms. Wang *et al.* (2017) clarified that a higher level of information reliability can potentially increase SCRM during market turbulence by accurately forecasting supply chain disruptions. Moreover, scholars have debated whether SCRM significantly impacts sustainable supply chains, including top management's strong commitment to upholding sustainability, maintaining a risk management culture (Reshad *et al.*, 2023) and encouraging innovation and learning in supply chain by reducing supply chain risks (Nisar *et al.*, 2022). However, this argument lacks empirical validation and previous discussions have been from the perspective of various small or large firms. Notably, some scholars have suggested that SCF could act as a direct driver of SCS by lowering the cost of financing, thereby increasing expenditures on sustainable practices (Wang *et al.*, 2023). Moreover, Jia *et al.* (2020) reviewed SCF papers and concluded that SCF could indirectly promote SCS by encouraging supply chain innovation through enhanced collaboration. Given the potentially comprehensive impact of SCF on SCS, we propose that SCF could also indirectly impact SCS through SCRM as a mediator, in addition to the direct SCF–SCS relationship. Essentially, SCRM acts as a bridge linking SCF with SCS to implement modifications and adjustment to insights. This facilitates the translation of reliable information to promote learning and innovative activities towards advancing SCS. However, the mediating effect of SCRM has not yet been empirically investigated.

Nevertheless, despite the widespread recognition and advantages of SCF, and the potential influence of SCRM within the context of SCS, there is a scarcity of literature validating triangular interconnections and the mediating effect of SCRM between these variables. This gap is particularly evident in the context of Chinese SME, which forms the background for our study. Furthermore, when viewed through the perspective of the dynamic capabilities view, the essential components of SCF (Rajaguru *et al.*, 2022), SCRM (El Baz and Ruel, 2021) and SCS (Chowdhury and Quaddus, 2021) are acknowledged as ordinary (lower-order) and dynamic (higher-order) capabilities within the supply chain domain. Based on the principles of dynamic capabilities view, the primary objective of this study is to discern the three-dimensional relationships between SCF, SCRM and SCS as well as the mediating effect of SCRM between SCF and SCS within the framework of SMEs.

This research makes several noteworthy contributions to existing literature. Firstly, it lays the foundation for future investigations that further apply dynamic capabilities view to explore the relationships between SCF and SCRM in the SCS literature. Secondly, this study delves more deeply into the profound implications of SCRM, thereby enriching its implications for achieving enhanced SCS and contributing to the SCS literature. Thirdly, the interrelationships among these variables were scrutinised by discerning the triangular connections between SCF, SCRM and SCS. Moreover, this study's findings provide SMEs with comprehensive insights that enable them to embrace and apply SCF and SCRM, thereby fostering enduring performance and advancing SCS. Finally, by using necessary condition analysis and partial least squares structural equation modelling (PLS-SEM) to dissect the intricate relationships within the research framework, this study addresses the aforementioned issues by gathering data from 319 Chinese SMEs.

2. Literature review

2.1 Dynamic capabilities view

The dynamic capabilities view expands on the resource-based view theory, which emphasises the characteristics of valuable, rare, inimitable and non-substitutable (Barney, 1991; McDougall *et al.*, 2022b). The dynamic capabilities concept was defined by Teece (2012, p. 1395) as "the company's ability to integrate, build and reconfigure internal and external resources/competencies to deal with and possibly adapt to rapidly changing business environments". In the context of dynamic capabilities view, replicability and imitability are determinant factors of an organisation's capabilities (Teece

et al., 1997). Dynamic capabilities view is thereby categorised into ordinary (lower-order) activities/capabilities and dynamic (higher-order) capabilities, where ordinary capabilities are easily replicable, and dynamic capabilities are difficult to replicate (Teece, 2014). Ordinary capabilities are represented by practical routines that can be broken into governance, administrative and operational capabilities (Schilke, 2014; Teece, 2014), whereas dynamic capabilities refer to the reconfiguration of ordinary capabilities through strategic insights and complex capacities (Schilke, 2014; Fainshmidt *et al.*, 2016). In addition to categorising dynamic capabilities view, scholars have identified its key processes to provide a deeper understanding. Pitelis *et al.* (2023) concurred with Teece (2007, 2014) that there are three core processes of dynamic capabilities view:

1 Sensing – capacity to recognise, evaluate, develop and analyse market prospects or threats; seizing – capacity to react to recognised opportunities or risks;
2 Allocation of resources to tackle opportunities and derive value from them; and
3 Reconfiguration (transformation)–continued renewal of organisational structures, business models and resources.

Therefore, the current study aims to offer a robust comprehension by using both the categorisation and core processes of dynamic capabilities view for grounding theoretical analysis.

This study investigates SCF from the perspective of dynamic capabilities view, in accordance with previous research which considered SCF as a type of capability or resource (Wang *et al.*, 2023; Rajaguru *et al.*, 2022). Rajaguru *et al.* (2022) identified SCF as encompassing activities and processes, such as account payables, receivables and payment processing, which facilitate supply chain operations, thereby categorising SCF as an ordinary capability. For instance, accounts payables and receivables enable organisations to smooth their capital flows and gather information on business, social and environmental requirements or risks. Wang *et al.* (2023) argued that SCF can also serve as a resource that can amplify capital. Therefore, regarding dynamic capabilities view, SCF can be considered as an ordinary capability that conducts operational routines.

Dynamic capabilities view is also a framework used by scholars to analyse how firms orchestrate their resources and capabilities to manage supply chain risks (Fan and Stevenson, 2018; Chowdhury and Quaddus, 2017). SCRM involves the synthetic management of essential operational competencies that cover risk identification, assessment, mitigation and control (El Baz and Ruel, 2021). These SCRM processes are performed as practical routines for organisations to manage and mitigate supply risks. For example, the routines of risk identification and risk assessment are applied to detect and estimate potential risks using risk-related information (Sun *et al.*, 2023). Risk mitigation practices endeavour to address risks (El Baz and Ruel, 2021). Finally, risk control involves continuous renewal practices through continuous monitoring of risks and adjusting measures, if necessary, at each step of SCRM (Kern *et al.*, 2012). In the current study, SCRM can sense risks according to the information gathered from SCF, risk identification and risk assessment, seize the opportunities to tackle the risk through RM and finally reconfigure the resources or capabilities to appropriately counter the risks. Therefore, SCRM exhibits practical components (SCRM practices) and strategically dynamic directions (overall-scope SCRM). The overall-scope SCRM illustrates the reconfiguration capacity attribute for the company, demonstrating that the company can reconfigure its ordinary capabilities (i.e. SCF, risk identification, risk assessment, risk mitigation and risk control) in response to risks and shifts in market demand or altered conditions (Gani *et al.*, 2023). Helfat and Winter (2011) argued that the line between ordinary and dynamic capabilities could be blurred by the existence of dual-purpose and multiple-variant capabilities. Thus, SCRM can be considered a dynamic capability consisting of ordinary capabilities (i.e. risk identification, risk assessment, risk mitigation and risk control) with both practical and strategic directions.

From the perspective of dynamic capabilities view, SCS practices are dynamic in nature and are applied to address dynamic stakeholder requirements around sustainability or to mitigate the risks that arise (Beske, 2012; Chowdhury and Quaddus, 2021). SCS is a process that dedicatedly reconfigures ordinary capabilities (e.g., from the perspectives of managing information, risks and capital flows) to satisfy stakeholder sustainability requirements through economic, environmental, social and operational considerations (Chowdhury and Quaddus, 2021). In the current study, SCS can sense the opportunities provided by SCF and SCRM (e.g., gathered information), seize opportunities (e.g., smooth cash flow and alleviate risks) and reconfigure the required assets and activities according to stakeholders' requirements. Therefore, SCS is considered a dynamic capability that can sense valuable information, seize opportunities and reconfigure and allocate ordinary capability (i.e. SCF) according to the dynamic sustainability requirements of stakeholders. Moreover, SCS primarily transfers stakeholders' requirements to the overall-scope SCRM (dynamic capability); that is, SCS initially reconfigures the overall-scope SCRM. Subsequently, the overall scope of SCRM is reconfigured and allocated to specific practical SCRM processes (risk identification, risk assessment, risk mitigation and risk control). Therefore, we argue that SCS reconfigures both ordinary capabilities (i.e. SCF) and dynamic capabilities (i.e. the overall-scope SCRM) in response to the sustainable requirements of stakeholders towards SCS improvement.

Thus, regarding the relationship between SCF, SCRM and SCS, SCF is considered an ordinary capability that gathers market information and assists SCS in achieving sustainable targets by supplementing capital. Moreover, practical SCRM processes exert the effects of ordinary capabilities, whereas the overall scope of SCRM presents the nature of dynamic capabilities in reconfiguring the routines of SCRM processes to address risks. Finally, SCS is a dynamic capability that ultimately reconfigures the routines of SCF and SCRM strategically towards establishing competitive advantages. Therefore, we propose a sequential relationship between these variables (SCF → SCRM → SCS) by theoretically grounding on dynamic capabilities view, which is a perspective that has not been previously explored.

2.2 Supply chain finance

Johnson and Templar (2011) defined SCF as a pivotal solution for addressing contemporary credit challenges, highlighting its contribution to improving an organisation's total financial performance and reducing the risks associated with supply

chain interruptions, both operationally and financially. SCF is considered a potent tool for significantly reducing firms' overall costs within a supply chain, particularly in terms of capital costs (Beka Be Nguema *et al.*, 2022). Notably, Gelsomino *et al.* (2016) identify a key gap in the literature, specifically the limited understanding of the impact of SCF adoption on financial performance. Further exploration of this gap is the primary focus of our study, as we aim to explore this aspect within the realm of SCS, encompassing financial and economic performance, particularly within the context of SMEs.

In the realm of SMEs, information asymmetry is a negative characteristic (Song *et al.*, 2018), because of their lower level of willingness to share information which exposes their weaknesses (Silva *et al.*, 2021). However, the weaknesses of SMEs cannot be easily hidden if they desire apply for SCF, as this requires providing business-related information to finance providers to check their qualifications (Moro *et al.*, 2015). In other words, SCF could be an effective approach for addressing information asymmetry in SMEs. Consequently, SCF research in the context of SMEs frequently relates to information from various perspectives. For instance, Song *et al.* (2020) conducted a series of case studies to analyse the possibility of SCF resolving information asymmetry. Similarly, Ali *et al.* (2018) stated that SCF not only facilitates SMEs' performance but also integrates it with digital trading, which improves connections through information sharing. In addition to information systems, Zhang *et al.* (2021) investigated how information systems integration and client interaction affect finance providers' decisions to offer SCF to SMEs. Previous research has recognised the importance of information for SMEs in performing SCF to achieve various outcomes. However, few empirical studies have highlighted the improvement of SCF in SCS in the context of SMEs, particularly in an age in which SCS is becoming increasingly important. Therefore, the present study aims to offer empirical insights based on the lens of dynamic capabilities view to explore the comprehensive mechanism, including but not limited to the information-related effect and towards identifying the relationship between SCF, SCRM and SCS to fill the research gap.

2.3 Supply chain risk management

SCRM has been addressed from various perspectives in the current literature. In this study, we define SCRM as recognising possible risk factors and introducing correct strategies to decrease the probability of risk in the supply chain (Foli *et al.*, 2022). Referring to the groundwork established by König and Spinler (2016) and El Baz and Ruel (2021), we identified four essential activities crucial to SCRM:

1 risk identification;
2 risk assessment;
3 risk mitigation; and
4 risk control.

Lima *et al.* (2020) provided a comprehensive review of SCRM literature, stressing that risk management in SMEs has both connections and differences compared to large firms. Some risk management practices suitable for large firms do not fit the requirements of SMEs, highlighting the necessity of investigating SCRM in SMEs. Compared to large firms, SMEs are more vulnerable to disruptions that can adversely affect their operations and supply chains. Given this vulnerability, scholars have suggested that SMEs should be more engaged in SCRM (Lima *et al.*, 2020). However, fully implementing SCRM practices may not be easy for resource-limited SMEs and selectively implementing SCRM would be more suitable for SMEs when considering cost performance. Thun *et al.* (2011) argued that SMEs with less cash flows and equity would not be able to cope with disruptions, leading to vulnerability. They emphasised the significance of proactive (i.e. preventive) SCRM strategies in alleviating the vulnerability by reducing the occurrence of risks. Recently, with the development of technology and an increasingly turbulent business landscape, the investigation of SCRM has continued, with a focus on IT technology adoption and comprehensive exploration. The necessity of adopting digital technologies to enhance the performance (e.g. risk management) of SMEs has been highlighted by some scholars (Papadopoulos *et al.*, 2020). Wong *et al.* (2022) concurred with Papadopoulos *et al.* (2020), proposing that implementing SCRM based on artificial intelligence encourages supply chains to dynamically respond to turbulent environments and mitigates potentially costly decision-making for SMEs. In addition, some scholars have endeavoured to provide a more detailed and deeper exploration of SCRM for SMEs in terms of flexibility, agility, robustness and resilience concerning proactive–reactive capabilities and demand-supply sides towards supply chain financing performance (Qiao and Zhao, 2023). However, while the exploration of SCRM is abundant, there is still limited empirical research that analyses the relationship between SCRM and SCS for SMEs. This study aims to fill this gap by detecting a potentially comprehensive relationship between them.

2.4 Supply chain sustainability

The definition of SCS in this context is derived from Chowdhury *et al.* (2023) and is consistent with Martins and Pato (2019) systematic review of SCS literature. SCS is defined as the management and coordination of resources, information and capital flows in a supply chain to meet the economic, environmental, social and operational goals of stakeholders and supply chain participants. Integrating sustainability into a supply chain is important for promoting an organisation by improving its brand image, economic performance and efficiency, while also positively benefiting the local community (Narayanan *et al.*, 2019). Chowdhury *et al.* (2023) presented a thorough method for evaluating supply chain performance with a focus on sustainability. This method has four interrelated components:

1 social;
2 economic;
3 environmental; and
4 Operational sustainability.

Aligning with the concept of the triple bottom line, which is the primary opinion on sustainability raised by Elkington (1998), SCS initially focused on the dimensions of society, environment and economics (Carter and Rogers, 2008; Carter and Liane Easton, 2011). However, following the classical categorisation of firm-based sustainability would not be

sufficient to validate SCS. For instance, identifying the economic sustainability merely through "financial" measurements (e.g. profit, cost and market share) would not be able to cover the satisfaction of stakeholders for production. Thus, some scholars argue that the operational aspect of sustainability is an additional vital component of SCS (e.g. product quality, conformance and reliability) (Varsei and Polyakovskiy, 2017; Kusi-Sarpong *et al.*, 2019; Chowdhury and Quaddus, 2021; Chowdhury *et al.*, 2023). In the current study, not only can SCF promote production by supplementing capital, but SCRM can also improve production by mitigating operation-related risks in the supply chain context. Therefore, the authors considered operational sustainability as a part of the SCS to provide more meaningful insights.

In the SME context, scholars argue that sustainability could be an effective approach for supporting SMEs' competitive edge (Yacob *et al.*, 2019). Consequently, research related to SCS has been conducted in multiple aspects to acquire a more thorough understanding of SMEs. Some scholars have explored the path to acquiring SCS. For example, Rupasinghe and Wijethilake (2021) pointed out that using lean techniques can help SMEs improve their manufacturing, quality management, energy efficiency and natural resource costs and increase employee engagement, thereby attaining SCS. However, some scholars have discussed the mechanisms by which SCS promotes SMEs' performance. For instance, some scholars revealed that SCS can improve the overall performance of SMEs (e.g., supplier performance, customer performance and operational performance) by addressing social concerns in the supply chain towards promoting effectiveness (Mani *et al.*, 2020)). Moreover, SCS can also contribute to SMEs' performance by processing the effect of factors (social pressure, green financial benefits and environmental commitment) to enhance circular economic capability, which may not directly affect SMEs' circular economic capability. Although scholars have explored SCS in the SME context from various dimensions, insufficient attention has been paid to empirical tests that elucidate the triangular relationship between SCF and SCRM. Thus, we conducted an empirical study to comprehend the relationship between SCF, SCRM and SCS from the lens of dynamic capabilities view, synthetically considering the four perspectives of SCS to address the research gap.

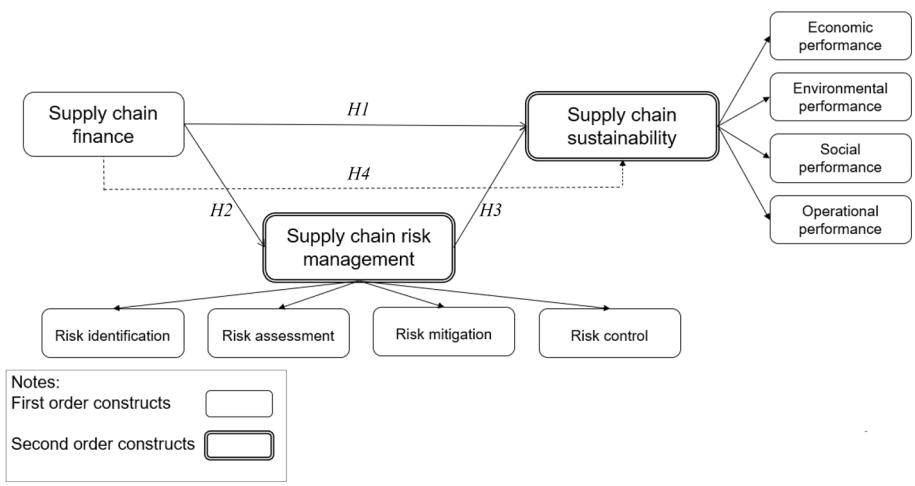
3. The research model and hypotheses development

Figure 1 depicts the research model for this study, which aims to explore the intricate relationships between SCF, SCRM and SCS.

3.1 Relations between supply chain finance and supply chain sustainability

SCF is a financing strategy based on actual transactions among supply chain members. It functions with the assurance of credit from the main company and provides cost-effective financing and easy accessibility (Liu *et al.*, 2020). Core firms play a crucial role in providing SCF to their supply chain partners to reduce potential financing expenses, lower overall cost outlays and enhance customer retention (Nigro *et al.*, 2021). Importantly, the cost-effectiveness of SCF can help decrease corporate resource consumption in line with sustainable development principles (Liu *et al.*, 2020). Moreover, SCF and SCS have similar goals and align thematically, both focusing on the concept of "doing well by doing good" (Toktaş-Palut, 2022). SCF prioritises the interests of all stakeholders by solely maximising individual interests. In other words, SCF represents not only self-interest, but also altruism. Consequently, supply chain participants consistently show readiness to engage in SCF, which improves the overall performance of the supply chain and helps members achieve SCS (Wang *et al.*, 2023). This viewpoint not only enhances the resilience of the supply chain (Gelsomino *et al.*, 2016) but also

Figure 1 Research model



Source: Developed for the research

promotes significant stakeholder dedication and aids in developing a sustainable network ecosystem (Bals, 2019). Particularly, SMEs struggle with insufficient liquidity and operating capital, which are worsened by economic downturns and fluctuations, resulting in corporate financing expenses. SCF is considered to be a reliable solution for SMEs to address these challenges. This facilitates the provision of working capital and reduces financial costs (Song *et al.*, 2018). Hence, SCS can benefit from SCF in SMEs. Therefore, the following hypothesis is proposed:

H1. In Chinese SMEs, SCF positively correlates with SCS.

3.2 Relations between supply chain finance and supply chain risk management

By enhancing SCRM capabilities, SCF effectively reduces financial and operational risks within the supply chain. According to Ali *et al.* (2020), SCF is one of the most effective methods for optimising working capital and mitigating financial risks in the supply chain. Furthermore, SCF's impact extends beyond financial benefits. The implementation of SCF can positively influence both physical and cash flows within the supply chain, thereby contributing to enhanced information reliability. Information quality can be significantly improved through the adoption of supply chain financing, leading to a reduction in market disruption risk (Wang *et al.*, 2017). In the traditional approach, financial institutions often provided financing for a firm's inventory and other resources, which inherently raised organisational risks (Ali *et al.*, 2020). However, Song *et al.* (2018) conducted a comparative analysis of SCF solutions offered by banks and financial service providers, demonstrating their ability to support SMEs in obtaining lower-risk credit to manage supply chain risk effectively. Hence, SCRM can be considered to benefit from SCF, and the following hypothesis is established:

H2. In Chinese SMEs, SCF positively correlates with SCRM.

3.3 Relations between supply chain risk management and supply chain sustainability

Effective SCRM is vital to ensure long-term industrial and organisational sustainability. Cooperation with customers and investment recovery are two core conditions to achieve SCS; however, such conditions can only be realised while taking high risks. In other words, SCRM is necessary to promote SCS by mitigating high-level risks (Silva *et al.*, 2021). Scholars also presented evidence that fostering supply chain innovation is an effective approach for SCRM to enhance SCS (Nisar *et al.* (2022). Florin (2022) elucidated that innovation entails significant uncertainty, implying potential unknown and unexpected risks. While innovativeness has the potential to reduce supply chain costs and enhance profit margins, contributing to the success of SCS (Bag *et al.*, 2020), it necessitates robust SCRM practices to mitigate the associated risk exposures. SMEs are typically marked not only by their susceptibility to supply chain risks (Foli *et al.*, 2022) but also by the difficulty in sustaining both innovation activities and innovation performance (Yıldız *et al.*, 2021). From the dynamic capabilities view perspective, the preference for innovation can be seen as the sustainable requirements of stakeholders, which may shift the focus of SCRM, thereby reconfiguring practical SCRM processes in response to stakeholders' requirements towards SCS enhancement. Consequently, SCRM has emerged as one of the most crucial enablers for SMEs to achieve SCS. Therefore, we argue that SCRM can promote SCS and establish the following hypothesis:

H3. In Chinese SMEs, SCRM positively correlates with SCS.

3.4 The mediating role of supply chain risk management

In regards to the dynamic capabilities view, both SCRM and SCS are identified as dynamic capabilities in the current study, whereas SCF is seen as an ordinary capability in a supply chain (Wang *et al.*, 2023; El Baz and Ruel, 2021). In particular, SCF is essential for improving information quality in a changing environment, which helps boost the credibility of supply chain participants and results in more effective SCRM (Wang *et al.*, 2017). SCF allows SMEs to build robust connections and cultivate mutual confidence in their supply chain collaborators. This improves the quality and effectiveness of knowledge acquisition, which is crucial for supply chain learning towards advancing SCRM (Liu *et al.*, 2023). In addition, Pan *et al.* (2021) proposed that improved implementation of SCF can greatly enhance innovation performance. By maintaining a steady cash flow and profitability through SCF, SMEs can efficiently use the resources already present in the supply chain network, which enhances their ability to innovate. The learning and innovation capabilities of SCRM are crucial for improving SCS. Therefore, we propose that SCRM's learning and innovation capabilities' towards SCS can be effectively promoted by the outcomes of SCF such as credibility, stable cash flow and profitability. The mediating effect of SCRMs can lead to higher SCS levels in a dynamic environment. Hence, the following hypothesis is proposed:

H4. In Chinese SMEs, SCRM plays a mediating role in the connection between SCF and SCS.

4. Research methodology

4.1 Sampling and data collection method

The primary focus of this investigation was SMEs because of their pivotal position in the economic landscape and the substantial challenges they face in attaining sustainable supply chain practices in the long term. Scholars have suggested that the selection of Chinese SMEs can be facilitated through the China Association of Small and Medium Enterprises, a recognised authority for SMEs in China, representing a diverse range of geographical locations and industries (Su *et al.*, 2019). This study followed Lim *et al.* (2022) recommendation that engaging a team of professional data collectors could improve data reliability and quality. These collectors distributed questionnaires to managers or owners across various departments in the SMEs of this association and individuals

with access to supply chain information. To ensure respondent anonymity, we applied a simple random sampling technique using this association's member list (Wong *et al.*, 2020). Moreover, a pre-test was conducted before conducting the pilot and full-scale tests. During this phase, two academics and two operations management practitioners reviewed the content to ensure validity. Adjustments were made to align with the context of Chinese SMEs, and any inconsistencies or technical jargons were revised based on their feedback. These inputs were instrumental in mitigating potential bias in the questionnaire. A pilot test was conducted with 30 participants to improve the research design and quality of the questionnaire items. Given that this research was conducted in China, it was necessary to ensure linguistic equivalence between English and Chinese by involving forward and backward translations (Koller *et al.*, 2012). Data were collected using a self-administered questionnaire, culminating in the acquisition of 319 valid responses as detailed in Tables 1 and 2. The response rate was 19.9%, with a total of 1, 600 questionnaires distributed. This exceeds the minimum requirement of 103 usable data sets specified by the G* Power program. Furthermore, this sample size adheres to the proposed "10 times rule", thereby complementing the PLS-SEM analysis and necessary condition analysis (Hair *et al.*, 2019).

This study carefully used and modified the measurement instruments used in previous research to assess four key constructs: SCF (Ali *et al.*, 2020), SCRM (El Baz and Ruel, 2021) and SCS (Chowdhury *et al.*, 2023). A seven-point Likert scale was used to evaluate these constructs, with the spectrum extending from "strongly agree" (7) to "strongly disagree" (1) to capture respondents' perceptions of the provided statements.

4.2 Integration of small and medium-sized enterprises and necessary condition analysis

The current study evaluates the use of variance-based SEM and necessary condition analysis within a selected statistical approach. The validation of the research model was done through using SmartPLS (version 4.0.9) for PLS-SEM and R software (version 3.3.2) for necessary condition analysis. Necessary condition analysis, proposed by Dul (2016) and further developed by Dul *et al.* (2020), is a new way to analyse data. The aim of necessary condition analysis is to assess the necessary-but-not-sufficient condition effects that exist between independent and dependent variables. In essence, this methodology demands that researchers and practitioners comprehend two key aspects: sufficiency factors are the factors that lead to an increased intention to use a technology; and necessity factors outline the requirements for this usage (Richter *et al.*, 2020). Therefore, to thoroughly analyse the relationship between SCF, SCRM and SCS, this study used both necessary condition analysis and PLS-SEM as research tools.

Table 1 Demographic profile

Characteristics		Frequency	(%)
Gender	Male	111	34.80
	Female	208	65.20
Age (years)	Below 26	4	1.25
	26–30	44	13.79
	31–35	187	58.62
	36–40	50	15.67
	41–45	16	5.02
	Above 45	18	5.64
The highest level of education	Secondary education or below	7	2.19
	Diploma/advance diploma	6	1.88
	Bachelor degree/ professional qualification	202	63.32
	Postgraduate (master/ doctorate)	104	32.60
Your job position	Junior management	68	21.32
	Middle management	167	52.35
	Senior/top management	84	26.33

Source: Developed for the research

Table 2 Organisational profile

Item		Frequency	%
Organisation age (years)	Less than 1	9	2.82
	1–2	6	1.88
	3–5	17	5.33
	6–10	66	20.69
	More than 10	221	69.28
Sales turnover (millions CNY)	Less than 5	5	1.57
	5–10	9	2.82
	11–20	13	4.08
	21–50	58	18.18
	51–150	71	22.26
	More than 150	163	51.10
Total property (millions CNY)	Less than 5	2	0.63
	5–10	8	2.51
	11–30	12	3.76
	31–80	47	14.73
	81–200	84	26.33
	More than 200	166	52.04
Ownership:	Foreign owned company	28	8.78
	State owned company	102	31.97
	Local private owned company	189	59.25

Source: Developed for the research

5. Results

5.1 Common method bias

Consistent with Podsakoff *et al.* (2003), similar to any self-reported data, consistency motives and social desirability could be potential sources for common method biases. This study used a range of remedies encompassing procedural and statistical approaches. Among the statistical remedies, the common latent method factor validation was used to assess 36 indicators from the research model (Podsakoff *et al.*, 2003, 2012). As shown in Table 3, the results indicate that common method bias is not a substantial concern. The average Ra^2 (0.656) surpassed the average Rb^2 (0.029), and most Rb^2 values were not significant. Consequently, common method

Table 3 Common method factor analysis

Latent construct	Indicators	Substantive factor loading (Ra)	Ra^2	Method factor loading (Rb)	Rb^2
EC	EC → EC1	0.881	0.776161***	−0.083	0.006889^{NS}
	EC → EC2	0.706	0.498436***	0.076	0.005776^{NS}
	EC → EC3	0.871	0.758641***	−0.017	0.000289^{NS}
	EC → EC4	0.832	0.692224***	0.027	0.000729^{NS}
EN	EN → EN1	0.905	0.819025***	−0.175	0.030625*
	EN → EN2	0.813	0.660969***	−0.018	0.000324^{NS}
	EN → EN3	0.888	0.788544***	−0.063	0.003969^{NS}
	EN → EN4	0.867	0.751689***	−0.149	0.022201^{NS}
	EN → EN5	0.413	0.170569***	0.395	0.156025***
OP	OP → OP1	0.968	0.937024***	−0.159	0.025281*
	OP → OP2	0.833	0.693889***	−0.052	0.002704 ^{NS}
	OP → OP3	0.956	0.913936***	−0.201	0.040401**
	OP → OP4	0.45	0.2025***	0.394	0.155236***
RA	RA → RA1	0.92	0.8464***	−0.089	0.007921^{NS}
	RA → RA2	0.888	0.788544***	−0.048	0.002304^{NS}
	RA → RA3	0.686	0.470596***	0.141	0.019881^{NS}
RC	RC → RC1	1.068	1.140624***	−0.322	0.103684**
	RC → RC2	0.627	0.393129***	0.193	0.037249^{NS}
	RC → RC3	0.771	0.594441***	0.034	0.001156^{NS}
	RC → RC4	0.793	0.628849***	0.072	0.005184^{NS}
RI	RI → RI1	0.863	0.744769***	−0.024	0.000576^{NS}
	RI → RI2	0.688	0.473344***	0.144	0.020736*
	RI → RI3	0.943	0.889249***	−0.117	0.013689^{NS}
RM	RM → RM1	0.855	0.731025***	−0.037	0.001369^{NS}
	RM → RM2	0.759	0.576081***	0.065	0.004225^{NS}
	RM → RM3	0.857	0.734449***	−0.028	0.000784^{NS}
SCF	SCF → SCF1	0.931	0.866761***	−0.199	0.039601**
	SCF → SCF2	0.799	0.638401***	0.046	0.002116^{NS}
	SCF → SCF3	0.746	0.556516***	0.125	0.015625^{NS}
SP	SP → SP1	0.782	0.611524***	0.002	0.000004^{NS}
	SP → SP2	0.804	0.646416***	0.03	0.0009^{NS}
	SP → SP3	0.857	0.734449***	−0.136	0.018496^{NS}
	SP → SP4	0.714	0.509796***	0.092	0.008464^{NS}
	Average:	0.793393939	0.656299697	0.013484848	0.029477848

Notes: *Significant at 5% level, $p < 0.05$; **Significant at 1% level, $p < 0.01$; ***Significant at 0.1% level, $p < 0.001$. NS insignificant
Source: Developed for the research

bias cannot be considered a major problem in the current study.

5.2 Inspecting the measurement model

Following the recommendation of Lee *et al.* (2024), this study evaluated reflectively measured constructs by ascertaining convergent validity, discriminant validity and internal consistency reliability. All composite reliability values range between 0.858 and 0.894 as shown in Table 4, meeting the required standards without significant concerns; these values surpass the minimum threshold of 0.70, as highlighted by Henseler *et al.* (2016). The average variance extracted and factor loadings were used to confirm convergent validity, as shown in Table 4. The factor loading value is required to be more than 0.7 to maintain convergent validity, and the average variance extracted value should be greater than the minimal threshold of 0.5 (Lee *et al.*, 2021). Consequently, the factor loading values ranged from 0.731–0.861 and met the required criteria. All the values of average variance extracted were between 0.601 and 0.693; therefore, there was further proof of convergent validation. It should be noted that discriminant validity was checked with the Fornell–Larcker criterion approach, and the results are shown in Table 5. These findings indicate that the study meets discriminant validity, as not all correlational coefficients exceed the square roots of the average variance extracted of each construct (Leong *et al.*, 2023). In general, this study appropriately demonstrates the internal consistency reliability, discriminant validity and convergent validity of the reflectively measured constructs, as illustrated by the findings in Tables 4 and 5.

5.3 Assessing the structural model

All the values of variance inflation factor values were from 1 to 2.955, well below the boundary of 5. This indicated the absence of a major multicollinearity problem (Hair *et al.*, 2011). In addition, the bootstrapping technique was used for the hypothesis tests which consisted of no sign changes and 5,000 sub-samples. As shown in Figure 2 and Table 6, the path

Table 4 Outcomes of internal consistency reliability and convergent validity testing

Constructs	Items	Loadings	Average variance extracted (AVE)	Composite reliability
EC			0.678	0.894
	EC1	0.798		
	EC2	0.779		
	EC3	0.861		
	EC4	0.853		
EN			0.601	0.883
	EN1	0.742		
	EN2	0.785		
	EN3	0.825		
	EN4	0.731		
	EN5	0.79		
OP			0.636	0.875
	OP1	0.825		
	OP2	0.786		
	OP3	0.769		
	OP4	0.81		
RA			0.693	0.871
	RA1	0.848		
	RA2	0.845		
	RA3	0.803		
RC			0.655	0.884
	RC1	0.767		
	RC2	0.803		
	RC3	0.807		
	RC4	0.859		
RI			0.692	0.871
	RI1	0.836		
	RI2	0.83		
	RI3	0.83		
RM			0.679	0.864
	RM1	0.826		
	RM2	0.823		
	RM3	0.823		
SCF			0.669	0.858
	SCF1	0.746		
	SCF2	0.843		
	SCF3	0.86		
SP			0.62	0.867
	SP1	0.788		
	SP2	0.822		
	SP3	0.74		
	SP4	0.796		

Source: Developed for the research

analysis results are presented. As depicted in Table 6, the relationships pertaining to SCF (*H1*: $p = 0.000$, $\beta = 0.221$; *H2*: $p = 0.000$, $\beta = 0.813$), SCRM (H3: $p = 0.000$, $\beta = 0.719$) and SCS are both significant and positive.

5.4 Analysing the mediating effects

A bootstrapping approach was used to establish the mediating role of SCRM (Nitzl *et al.*, 2016; Lee *et al.*, 2018). The mediating effect results are summarised in Table 7 and graphically presented in Figure 2 for testing H4. The results indicate that SCRM serves as a mediator in the association between SCF and SCS, where SCF contributes a significantly direct effect on SCS ($p < 0.001$, $\beta = 0.221$). In addition, the indirect effect of SCF ($p < 0.001$, $\beta = 0.585$) on SCS through the mediator variable SCRM is significant. Hence, we can infer that the relationship between the dependent variable (SCF) and independent variable (SCS) is partly mediated by SCRM.

Table 5 Fornell–Larcker criterion

Item	EC	EN	OP	RA	RC	RI	RM	SCF	SP
EC	*0.824*								
EN	0.732	*0.775*							
OP	0.738	0.765	*0.798*						
RA	0.701	0.739	0.747	*0.832*					
RC	0.796	0.768	0.756	0.805	*0.81*				
RI	0.725	0.717	0.776	0.776	0.799	*0.832*			
RM	0.729	0.736	0.782	0.751	0.803	0.816	*0.824*		
SCF	0.729	0.698	0.758	0.708	0.777	0.704	0.794	*0.818*	
SP	0.69	0.728	0.757	0.718	0.701	0.709	0.718	0.699	*0.787*

Notes: The square root of the average variance extracted are the diagonal elements (in italics)
Source: Developed for the research

5.5 The effect size and predictive relevance

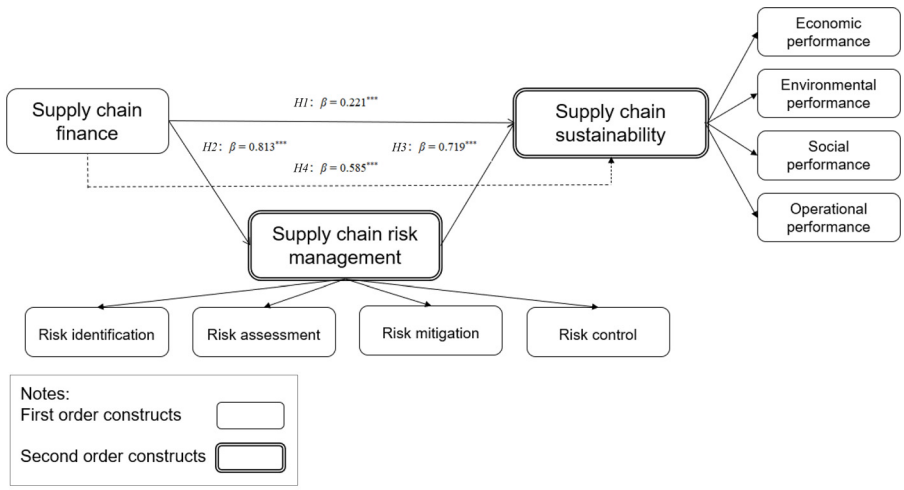
Table 8 presents the performance metrics of the structural model, including predictive relevance (measured by Q^2), predictive accuracy (indicated by R^2) and effect size (quantified by f^2). These metrics exert a crucial effect in assessing the ability of independent variables to predict dependent variables in a research model (Hair *et al.*, 2019). In terms of these parameters, Table 8 illustrates a range of effect size values, spanning from a small effect (0.094) to a highly significant effect (1.955). Furthermore, Table 8 demonstrates a substantial level of predictive relevance and accuracy concerning the endogenous constructs of SCRM ($R^2 = 0.662$ and $Q^2 = 0.552$) and SCS (with $R^2 = 0.825$ and $Q^2 = 0.653$).

5.6 Necessary condition analysis

By following the suggestion of Dul *et al.* (2020), we perform necessary condition analysis using R software (version 3.3.2). As Sharma *et al.* (2022) suggested, it is appropriate to assess the mediating effect of a particular "necessary-but-not-sufficient condition" by using necessary condition analysis. Moreover, to provide a complete analysis, Duarte *et al.* (2022) argued that it was essential to assess the dimensions of higher-order constructs (i.e. the dimensions of risk assessment, risk identification, risk control and risk mitigation within SCRM).

Initially, the scatter plots depicted in Figure 3 illustrate a phenomenon known as the "necessary-but-not-sufficient condition". This phenomenon can be acquired from the CE-FDH line and is recognised by the vacant area situated in the top-left corner of the scatter plots. Importantly, Vis and Dul (2018) explained that for analysing survey data using a seven-point Likert scale, the CE-FDH line is considered suitable. Another pivotal aspect involves the analysis of statistical

Figure 2 Results of hypotheses testing



Source: Developed for the research

Table 6 The structural model examination result

Relationship	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics (|O/STDEV|)	Confidence interval	*p*-Values	Remark
SCF → SCRM	0.813	0.813	0.028	28.595	(0.751, 0.862)	0	Supported
SCF → SCS	0.221	0.22	0.062	3.569	(0.101, 0.342)	0	Supported
SCRM → SCS	0.719	0.719	0.059	12.215	(0.599, 0.830)	0	Supported

Source: Developed for the research

Table 7 The outcome of the mediating effect examination

Hypothesis	IV	MV	DV	Direct effect	Indirect effect of IV on DV	SE of indirect effect	BI of indirect effect [2.5%; 97.5%]	Remark
H4	SCF	SCRM	SCS	0.221***	0.585***	0.053	0.475–0.684	Partially mediating

Notes: $p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$; NS: insignificant; IV: independent variable; MV: mediating variable; DV: dependent variable; SE: standard error; BI: confidence intervals bias corrected
Source: Developed for the research

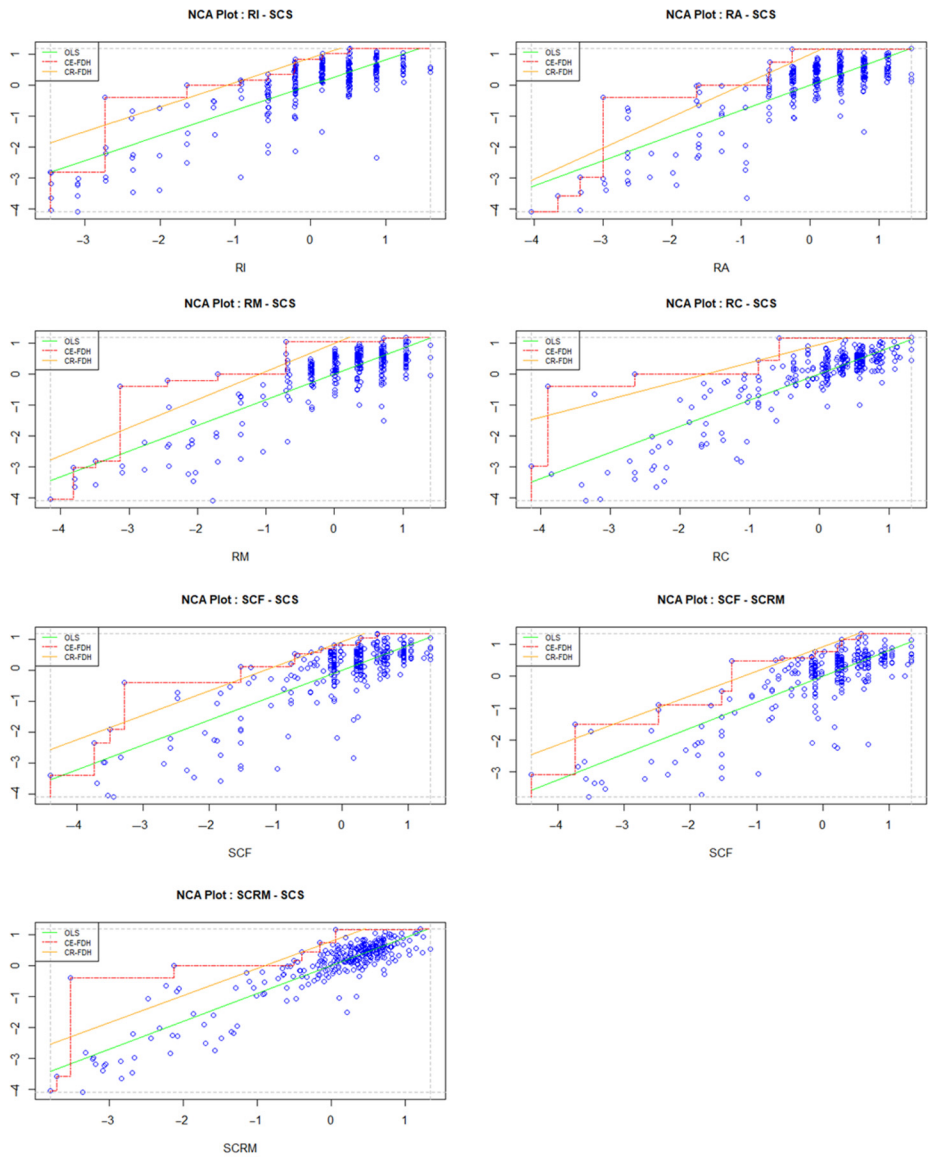
Table 8 Predictive accuracy, predictive relevance and effect size

Endogenous constructs	R^2	Q^2	Exogenous constructs	Effect size f^2
SCS	0.825	0.653	SCF	0.094
			SCRM	0.999
SCRM	0.662	0.552	SCF	1.955

Source: Developed for the research

significance and the necessary effect size (d). This analysis aids in determining whether a specific variable is a necessary condition for a phenomenon (Dul *et al.*, 2020). As shown in Table 9, all the outcomes exhibit effect sizes (d) ranging from 0.184 to 0.292. These values signify medium ($0.1 \leq d < 0.3$) effects and they are statistically significant (p-value < 0.001). Finally, as shown in Table 10, the bottleneck table provides a crucial threshold for SCF, SCRM, risk assessment, risk identification, risk mitigation and risk control, which are essential for attaining a favourable level of SCS. This table details the specific SCF levels necessary to achieve the target SCRM level. For instance, when detecting the variables for a 100% SCS level, SCF (86.1%) is less pivotal than SCRM (97.4%). Among the SCRM dimensions, risk assessment (100%) had the highest significance, followed by risk control (99.9%), risk mitigation (93.6%) and risk identification (78.8%). Overall, the results underscore the congruence between our data and the "necessary-but-not-sufficient" connections that exist within SCF, SCRM, risk assessment, risk identification, risk control and risk mitigation in relation to

Figure 3 Scatter plots



Source: Developed for the research

SCS. Furthermore, the data provide evidence for the interconnectedness between SCF and SCRM.

6. Discussion and implications

The current research aims at achieving SCS towards promoting long-term supply chain performance by investigating the connections among SCF, SCRM and SCS, with a specific focus on Chinese SMEs. We test our model using two approaches: necessary condition analysis and PLS-SEM. This study acknowledges the presence of a significant positive correlation between SCF-SCRM, SCF-SCS and SCRM-SCS. Additionally, SCRM plays a role as a partial mediator of the connection between SCF and SCS.

This study presents evidence highlighting *H1*, *H2* and *H3*, which demonstrate the vital roles played by SCF and SCRM in advancing SCS. In particular, our findings support the results of Wang *et al.* (2023) that SCF has the potential to improve SCS. SCF is characterised by economic efficiency and shared goals that ultimately help build a sustainable network ecosystem by encouraging stakeholder participation (Bals, 2019; Liu *et al.*, 2020). Moreover, our study demonstrated that SCF can improve SCRM and promote SCS development. This finding aligns with the assertion of Ali *et al.* (2020), who validated the presence of a positive correlation between SCF and SCRM. By using SCF, the possibility of market disruption can be reduced by improving information quality (Wang *et al.*, 2017). Furthermore, the PLS-SEM analysis showed a significantly positive association between SCRM and SCS. This finding is consistent with research conducted by Silva *et al.* (2021), who emphasised the influence of SCRM on the reduction of risks and uncertainty in SCS through increased

Table 9 Necessary condition analysis effect sizes

	CE-FDH of SCS		
Construct	**Effect size(d)**	**Conditional inefficiency**	**Outcome inefficiency**
RA	0.292***	0	0
RC	0.184***	0.055	21.168
RI	0.235***	21.216	24.355
RM	0.273***	6.412	1.1
SCF	0.287***	13.941	13.259
SCRM	0.215***	2.632	1.1
CE-FDH of SCRM			
SCF	0.345***	13.19	13.823

Notes: All the CE-FDH effect sizes are significant at $^{***}p < 0.001$ level (see effect size column). Accuracy of CE-FDH line is not mentioned in the table as it is always 100% by design
Source: Developed for the research

Table 10 Bottleneck table (percentages)

	Bottleneck SCS					
Percentage(%)	**SCF**	**RA**	**RC**	**RI**	**RM**	**SCRM**
0.00	NN	NN	NN	NN	NN	NN
10.00	NN	12.8	NN	NN	6	5.2
20.00	11.4	12.8	NN	NN	6	5.2
30.00	11.4	18.8	4.4	14.2	18.4	5.2
40.00	15.7	18.8	4.4	14.2	18.4	5.2
50.00	19.4	18.8	4.4	14.2	18.4	5.2
60.00	19.4	18.8	4.4	14.2	18.4	5.2
70.00	19.4	18.8	4.4	14.2	18.4	5.2
80.00	63.2	62.3	59.8	50	61.9	64.1
90.00	71.2	62.8	65.2	64.2	61.9	70.9
100.00	86.1	100	99.9	78.8	93.6	97.4
Bottleneck SCRM						
0.00	NN					
10.00	NN					
20.00	11.4					
30.00	11.4					
40.00	11.4					
50.00	33.4					
60.00	50					
70.00	52.8					
80.00	52.8					
90.00	80.6					
100.00	86.8					

Note: NN: not necessary
Source: Developed for the research

investments and innovations. From the perspective of dynamic capabilities view, this study concurs with previous research, indicating that ordinary capabilities can contribute to dynamic capabilities (Rajaguru *et al.*, 2022). In addition, the findings empirically validate the proposition that third-order (or even higher-order) dynamic capabilities (SCS) can reconfigure lower-level dynamic capabilities (SCRM) (Schilke, 2014). Overall, our findings are generally consistent with past research on SCF with SCRM and SCF with SCS, supporting the idea that SCRM helps enhance SCS effectiveness. In addition, *H4* is supported, providing evidence that SCRM acts as a partial mediator in the relationship between SCF and SCS. This result supports recent studies by Rajaguru *et al.* (2022) and El Baz and Ruel (2021) that underscore SCF as an ordinary capability and SCRM as a dynamic capability in the supply chain environment. SCF exerts a crucial impact in building credibility, maintaining consistent cash flow and promoting profitability, which helps improve SCRM's ability to learn and innovate supply chains, ultimately supporting SCS (Liu *et al.*, 2023; Lu *et al.*, 2023). Based on these foundational perspectives, we provide novel insights into the existence of comprehensive relationships in which SCF not only directly promotes SCS, but also exerts an indirect influence through the mediating mechanism of SCRM.

Furthermore, we used necessary condition analysis along with PLS-SEM, aligning with the recommendations of Richter *et al.* (2020) and Dul (2016), who advised the integration of complementary methodologies to yield supplementary insights. Analyses of the connections between SCS (with independent variables SCF and SCRM) and SCRM (with independent variable SCF) using both PLS-SEM and necessary condition analysis yielded statistically significant results, as indicated in Tables 6 and 9. The interpretive approach proposed by Richter *et al.* (2020) highlights a consistent trend: an elevation in the exogenous construct, on average, corresponds to an increase in the outcome; however, it is important to note a specific threshold (refer to Table 10) of the exogenous construct that is pivotal for the manifestation of the outcome. Furthermore, we harnessed necessary condition analysis to identify the essential conditions for the four dimensions of SCRM, all of which exhibited significant correlations with SCS (Table 9). Among the dimensions of SCRM, heightened attention should be directed towards risk assessment, risk control and risk mitigation if the target achieves the highest SCRM level (100%). A notable observation from the bottleneck table (Table 10) is that striving for a top-tier outcome for SCS and SCRM necessitates a dramatic increase in the significance of SCF concerning SCS in the range of 80% to 100%, as well as SCF concerning SCRM in the range of 60% to 100%. Overall, the PLS-SEM and necessary condition analysis results show that both SCF and SCRM can positively influence SCS and are necessary conditions for advancing SCS.

6.1 Theoretical implications

The current study enhances the literature regarding dynamic capabilities view in the triangular relationships of SCF, SCRM and SCS. Firstly, this research expands the utilisation of the dynamic capabilities view by discussing the dynamic capabilities view hierarchy (i.e. ordinary and dynamic capabilities) and processes (i.e. sensing, seizing and reconfiguration) to provide a more profound understanding of the factors and their relationships. This study focuses on these dynamic capabilities view processes to identify SCF and SCRM practices (risk identification, risk assessment, risk mitigation and risk control) as ordinary capabilities and recognises the overall scope of SCRM and SCS as dynamic capabilities. Secondly, the current research provides a deeper understanding of the boundary fuzziness of dynamic capabilities view and argues that SCRM could exert the effect of both ordinary and dynamic capabilities. This study proposes that the SCRM presented by the practical routines of risk

identification, risk assessment, risk mitigation and risk control could be considered ordinary capabilities, while SCRM could also reconfigure practical routines in strategic directions, thus being seen as a dynamic capability. Thirdly, this research enriches the dynamic capabilities view by identifying the effect of dual reconfiguration on ordinary capabilities and other dynamic capabilities, indicating the potential of third-order (or even higher-order) dynamic capabilities. This study further shows that SCS reconfigures SCRM, in which both SCS and SCRM are dynamic capabilities. This provides the potential for the dynamic capabilities view to observe the dual-reconfiguration effect of dynamic capabilities on both ordinary and other-order dynamic capabilities. Fourth, this study explored the triangular relationships incorporating SCF, SCRM and SCS. This research also provides SMEs with a comprehensive understanding of SCF and SCRM adoption, thereby encouraging their SCS performance. Finally, we adopt a multi-method approach by combining PLS-SEM and necessary condition analysis to enhance our understanding of the subject. Unlike previous studies primarily relying on sufficiency logic using methods like PLS-SEM and artificial neural networks, necessary condition analysis operates on necessary logic. This implies that necessary condition analysis emphasises the significance of "must-have" variables. Thus, the authors theoretically validated both sufficiency and necessary logic analyses in the research model, reinforcing this novel theoretical comprehension with empirical substantiation.

6.2 Managerial implications

The findings of this research provide significant insights from a managerial standpoint for chief executive officers, managers and stakeholders seeking to enhance long-term performance through SCS within SMEs. Viewed through the lens of dynamic capabilities view, SCF emerges as an ordinary capability that can contribute to the dynamic capabilities of SCRM and SCS; SCRM can promote SCS by sensing reliable information and seizing opportunities for SMEs. Thus, SMEs can use SCF and SCRM or either one to improve their SCS performance. However, considering that SMEs may lack the substantial resources to fully implement SCF and SCRM practices, they must strategically allocate their limited resources in the most effective and efficient manner. Therefore, we propose three suggestions to address this issue. Firstly, both the necessary condition analysis and PLS-SEM results underline the significance of SCF and SCRM in amplifying SCS. However, only SCF plays the role of an ordinary capability and may have the unique potential to contribute to both SCRM and SCS. SCF promotes SCRM and SCS by providing low-cost financial resources and reliable information. Therefore, SMEs should prioritise SCF, as it offers an exclusive dual advantage that addresses both financial needs and information reliability. Secondly, achieving the utmost level might face scrutiny for its cost performance. Although SMEs are primarily suggested to carry out SCF, the most efficient cost performance situation is targeting 50% level SCRM and SCS which both require only 33.4% and 19.4% necessity of SCF, as shown in the necessary condition analysis (Table 10). Finally, for firms that would like to execute SCRM besides carrying on SCF to further ensure SCS, the necessary condition analysis results indicate that pursuing no more than 70% SCS requires less than 20% SCRM practices (risk identification, risk assessment, risk mitigation and risk control). In other words, it is suggested that medium-level SCRM practices be performed to maintain SCS, as they present outstanding cost performance.

7. Conclusion, limitations and future research

In summary, this research presents a research framework for SCF, SCRM and SCS in Chinese SMEs. The results confirmed a significantly positive association between SCF, SCRM and SCS in the triangular construct. Furthermore, the findings demonstrate that SCRM serves as a partial mediator linking SCF and SCS.

As with any investigation endeavour, future studies should strive to tackle several inherent limitations identified in this research. It is crucial to emphasise that these findings are based on data obtained from the SMEs located in China. However, diverse organisational and geographical contexts could result in different outcomes. To enrich our understanding, future research should encompass global geographical regions and diverse organisational contexts, including factors such as category and size. Secondly, this study's cross-sectional design evaluated variable relationships in a single instance, limiting the insights to a specific time point. A longitudinal approach could be a potential method for future research to unravel the evolving nature of SCS and SCRM phenomena by applying a natural resource-based view and dynamic capabilities from a sustainable SC perspective (McDougall *et al.*, 2022a). Thirdly, value chain analysis is another insightful methodology suggested for future SCRM research. It identifies value-adding activities and highlights waste in supply chains (Fearne and Wagner, 2022). Finally, to achieve deeper insights, using a mixed method or qualitative research is recommended. This study primarily used a self-administered survey for quantitative analysis; however, a more comprehensive understanding can be derived from mixed or qualitative approaches.

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